Practice Problems: AP Precalc Unit 1

Name: _____ Date: _____

Unit 1: Polynomial and rational functions

Rates of change

W1. Find the average rate of change of $f(x) = 3x - 7$ on $[2, 6]$.

W2. Find the average rate of change of $f(x) = x^2$ on $[0, 2]$ and on $[2, 4]$. Which is bigger, and what does that say about the graph?

1. Use the table.

x	0	1	2	3	4
$f(x)$	1	4	9	16	25

- (a) Find the change in f over each step of 1.
- (b) Are the rates of change increasing or decreasing? Is f concave up or down?
- (c) What kind of function is it, and how do you know?

2. $f(x) = -x^2 + 6x$. Find the average rate of change on $[0, 2]$, $[2, 4]$ and $[4, 6]$. Describe what's happening to the rate of change.

3. A ball's height is $h(t) = -16t^2 + 64t$ feet after t seconds. Find the average rate of change on $[1, 3]$ and explain what the answer means (and doesn't mean).

4. ★ A function is increasing and concave down on an interval. Describe what the rate of change is doing, and give a real-world example.

Polynomials

W3. Find the zeros of $f(x) = (x-2)(x+5)$.

W4. For $f(x) = -2x^3+x-4$, give the degree, the leading coefficient and the end behavior.

5. $f(x) = (x+2)(x-1)^2(x-3)$.

- (a) List the zeros and their multiplicities. Where does the graph cross, and where does it bounce?
- (b) What is the degree? What is the end behavior?
- (c) Find the y-intercept.

6. Find all zeros, real and complex, of $x^3 - x^2 + 4x - 4$. (Hint: grouping.)

7. ★ A degree-3 polynomial with real coefficients and leading coefficient 1 has zeros 2 and $1 + i$. Find the third zero and write the polynomial in standard form.

8. Is each function even, odd or neither? $f(x) = x^4 - 3x^2,\ g(x) = x^3 + x,\ h(x) = x^3 + 1$

9. Use limit notation to describe the end behavior of $p(x) = -3x^5 + 2x^2 - 1$.

Rational functions

W5. For $f(x) = \dfrac{1}{x-3}$, give the vertical asymptote and the horizontal asymptote.

10. $r(x) = \dfrac{2x^2 - 8}{x^2 - x - 6}$. Find the zeros, holes, vertical asymptotes, horizontal asymptote and y-intercept.

11. $r(x) = \dfrac{x^2 + 1}{x - 1}$. Find the vertical asymptote and the slant asymptote.

12. Find $\lim\limits_{x \to \infty} \dfrac{3x + 1}{x^2 + 2}$ and $\lim\limits_{x \to \infty} \dfrac{6x^2 - x}{2x^2 + 5}$.

13. ★ Solve $\dfrac{x - 2}{x + 1} \geq 0$. (Make a sign chart.)

Equivalent forms

14. Divide: $(2x^3 + 3x^2 - x + 5) \div (x + 2)$.

15. Use the binomial theorem or Pascal's triangle to expand $(x+2)^4$.

Transformations

W6. Describe how $g(x) = f(x-3)+2$ relates to $f(x)$.

16. $f(x) = x^2$ and $g(x) = -2f(x+1) - 4$. Describe each transformation in order, and give the vertex of g.

17. ★ The point $(4, 6)$ is on the graph of f. What point must be on $g(x) = 3f(2x) - 1$?

Modeling

18. ★ A quadratic model passes through $(0, 5)$, $(1, 8)$ and $(2, 15)$. Find it, and use it to predict the output at $x = 5$.

Answer Key

- **W1.** 3
- **W2.** 2 on $[0, 2]$ and 6 on $[2, 4]$. The rate of change is increasing, so the graph is concave up.
- **1.** (a) 3, 5, 7, 9. (b) Increasing, so concave up. (c) Quadratic: the second differences are constant (all 2). It's $(x + 1)^2$.
- **2.** $f(0) = 0$, $f(2) = 8$, $f(4) = 8$, $f(6) = 0$, so the rates are 4, 0, -4. The rate of change is decreasing, so the graph is concave down (it rises, peaks, then falls).
- **3.** $h(1) = 48$, $h(3) = 48$, so the average rate is 0 ft/s. The ball ended at the same height it started on that interval. It doesn't mean the ball stopped: it went up and came back down.
- **4.** The rate of change is positive but getting smaller: still going up, but more slowly. Examples: a plant's height as it matures, or a car's speed as it approaches the speed limit.
- **W3.** $x = 2$ and $x = -5$
- **W4.** Degree 3, leading coefficient -2. Up on the left, down on the right: $x \to -\infty$ gives $f \to \infty$, and $x \to \infty$ gives $f \to -\infty$.
- **5.** (a) -2 (mult. 1, crosses), 1 (mult. 2, bounces), 3 (mult. 1, crosses). (b) Degree 4, leading coefficient +1, so up on both ends. (c) $f(0) = (2)(1)(-3) = -6$
- **6.** $(x - 1)(x^2 + 4)$, so $x = 1$, $2i$, $-2i$
- **7.** The third zero is $1 - i$ (conjugate pair). $(x - 2)(x^2 - 2x + 2) = x^3 - 4x^2 + 6x - 4$
- **8.** f even, g odd, h neither
- **9.** $\lim_{x\to\infty} p(x) = -\infty$ and $\lim_{x\to-\infty} p(x) = \infty$
- **W5.** Vertical asymptote $x = 3$, horizontal asymptote $y = 0$
- **10.** $r(x) = \dfrac{2(x-2)(x+2)}{(x-3)(x+2)}$. Zero $x = 2$. Hole at $x = -2$, at height $\frac{8}{5}$: the point $\left(-2, \frac{8}{5}\right)$. Vertical asymptote $x = 3$. Horizontal asymptote $y = 2$. y-intercept $\frac{4}{3}$.
- **11.** Vertical asymptote $x = 1$. Long division gives $x + 1 + \dfrac{2}{x-1}$, so the slant asymptote is $y = x + 1$.
- **12.** 0 and 3
- **13.** $x < -1$ or $x \geq 2$ (-1 is excluded because it makes the denominator 0)
- **14.** $2x^2 - x + 1$ with remainder 3: $2x^2 - x + 1 + \dfrac{3}{x+2}$
- **15.** $x^4 + 8x^3 + 24x^2 + 32x + 16$
- **W6.** Shift right 3 and up 2
- **16.** Shift left 1, stretch vertically by 2, reflect over the x-axis, shift down 4. Vertex $(-1, -4)$, opening down.
- **17.** Need $2x = 4$, so $x = 2$, and $y = 3(6) - 1 = 17$. The point is $(2, 17)$.
- **18.** $c = 5$, $a + b = 3$, $4a + 2b = 10$, so $a = 2$ and $b = 1$: $f(x) = 2x^2 + x + 5$, and $f(5) = 60$.